

PHENIKAA UNIVERSITY
FACULTY OF ELECTRICAL AND ELECTRONIC ENGINEERING

ADVANCED DEEP LEARNING COURSE

FACIAL EMOTION RECOGNITION

Midterm Report

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ADVANCED DEEP LEARNING COURSE REPORT

Abstract

Facial emotion recognition (FER) from still images supports human–computer interaction, driver monitoring and affective computing, but it remains difficult on small, in-the-wild faces. In this midterm study we build a clean and exactly balanced seven-class dataset of 53,088 grayscale 48×48 face images from four sources (a FER2013-style local set, RAF-DB, AffectNet and CK+), remove near-duplicate images that leaked across the training, validation and test splits, and split CK+ by subject. We design EmotionCNN, a 4.7 M-parameter convolutional network with batch normalization, dropout and dilated convolutions trained from scratch, and compare it with two ImageNet-pretrained networks fine-tuned end to end, ResNet18 and MobileNetV2. Every model is trained with three random seeds and evaluated once on a held-out test set of 1,351 images. EmotionCNN reaches $68.0 \pm 1.0\%$ accuracy (macro-F1 0.679), on par with ResNet18 ($68.3 \pm 1.0\%$) and MobileNetV2 ($67.8 \pm 0.9\%$), while needing only 13.5 ms per image on a laptop CPU. Averaging the nine trained models raises the test accuracy to 71.8%. Accuracy ranges from 91% on laboratory-posed CK+ faces to 62% on the noisy FER2013-style images, and fear is the hardest emotion (F1 0.52), mostly confused with surprise. These results suggest that label noise, more than model capacity, now limits performance, which motivates relabeling with FER+ annotations, Grad-CAM analysis and a real-time webcam application in the final report.

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Chapter 1. INTRODUCTION

1.1. Project Definition

Facial expressions are one of the most direct signals of a person's emotional state. Recognizing them automatically enables applications such as adaptive human–computer interfaces, driver drowsiness and distraction monitoring, e-learning engagement analysis and mental-health screening. The task is usually framed as classifying a face image into the seven basic emotions: angry, disgust, fear, happy, neutral, sad and surprise.

Although deep convolutional networks have made FER practical, faces captured *in the wild* vary strongly in pose, illumination, occlusion, age and image quality. Public datasets also differ widely: laboratory datasets such as CK+ contain posed, frontal expressions, while web-collected datasets such as FER2013 contain small 48×48 grayscale images whose labels are known to be noisy. A reliable FER system therefore depends as much on careful data preparation and honest evaluation as on the network architecture.

1.2. Project Objective

The midterm phase of the project focuses on building a sound baseline before the real-time application of the final phase. Its objectives are:

- to build a clean, exactly balanced and leakage-free seven-class dataset from several public sources;
- to design and train a convolutional network from scratch (EmotionCNN) using the techniques of the course: batch normalization, dropout and dilated convolutions;
- to compare it with two transfer-learning baselines, ResNet18 and MobileNetV2, under one reproducible protocol: three random seeds, a bootstrap confidence interval, and a test set that is used only once per final model;
- to analyze the remaining errors per emotion and per image source, in order to decide what the final phase should improve.

Chapter 2. LITERATURE REVIEW

2.1. Project Background

2.1.1. Convolutional Neural Networks

A convolutional layer slides a set of learnable kernels over its input and computes, at every position, the weighted sum of the covered values [6], [7]. For an input x with C channels and the k -th kernel W_k of size $K \times K$, the output feature map is:

$$y_k(i, j) = \sum_c \sum_m \sum_n W_k(m, n, c) \cdot x(i + m, j + n, c) + b_k \quad (1)$$

Weight sharing makes the layer translation-equivariant and far cheaper than a fully connected layer. Each convolution is followed by the non-linearity ReLU and, periodically, by max pooling, which halves the spatial resolution and enlarges the receptive field:

$$\text{ReLU}(z) = \max(0, z) \quad (2)$$

2.1.2. Batch Normalization and Dropout

Batch normalization [8] normalizes every channel with the mean μ_B and variance σ_B^2 of the mini-batch and then rescales it with learnable parameters γ and β . It stabilizes and speeds up training and allows larger learning rates:

$$\hat{x} = (x - \mu_B) / \sqrt{(\sigma_B^2 + \epsilon)}, \quad y = \gamma \cdot \hat{x} + \beta \quad (3)$$

Dropout [9] randomly sets activations to zero during training with probability p , which prevents co-adaptation of neurons and acts as an ensemble of thinned networks. Spatial dropout (Dropout2d) drops whole feature maps, which suits convolutional layers:

$$\tilde{h} = (m \odot h) / (1 - p), \quad m \sim \text{Bernoulli}(1 - p) \quad (4)$$

2.1.3. Dilated Convolution

A dilated convolution [10] inserts gaps of size r between the kernel taps. With the same number of weights it covers a $(K + (K - 1)(r - 1))$ -wide region, so the receptive field grows without further pooling and without losing resolution:

$$y(i, j) = \sum_m \sum_n W(m, n) \cdot x(i + r \cdot m, j + r \cdot n) \quad (5)$$

2.1.4. Residual and Efficient Architectures

ResNet [11] learns a residual function F on top of an identity shortcut, which keeps gradients flowing through very deep networks:

$$y = F(x, \{W_i\}) + x \quad (6)$$

MobileNetV2 [12] replaces standard convolutions with depthwise separable convolutions inside inverted residual blocks with linear bottlenecks. For a $K \times K$ kernel and N output channels, the computational cost relative to a standard convolution is:

$$\text{cost ratio} = 1/N + 1/K^2 \quad (7)$$

so a 3×3 depthwise separable layer is roughly 8–9 times cheaper. This makes MobileNetV2 a natural candidate for real-time applications on a CPU.

2.1.5. Transfer Learning and Training Objective

Networks pretrained on ImageNet [13] already contain generic edge, texture and shape detectors; fine-tuning them on a target task usually converges faster than training from scratch. The network outputs logits z , turned into class probabilities by the softmax function, and is trained with cross-entropy against label-smoothed targets q [14], which discourages over-confident predictions:

$$p_k = \exp(z_k) / \sum_j \exp(z_j) \quad (8)$$

$$L = - \sum_k q_k \cdot \log p_k, \quad q_k = (1 - \varepsilon) \cdot 1[k = y] + \varepsilon / K \quad (9)$$

We optimize with AdamW [15], which decouples weight decay from the gradient update, and a cosine learning-rate schedule [16] after a short linear warm-up:

$$\eta_t = \eta_{\min} + \frac{1}{2} (\eta_{\max} - \eta_{\min}) (1 + \cos(\pi t / T)) \quad (10)$$

2.2. Related Works

Early FER systems relied on hand-crafted descriptors such as local binary patterns [22], [25] or histograms of oriented gradients [23], followed by a support vector machine. They work well on posed laboratory data such as CK+ [4], where accuracies above 95% are commonly reported, but degrade on spontaneous, in-the-wild faces.

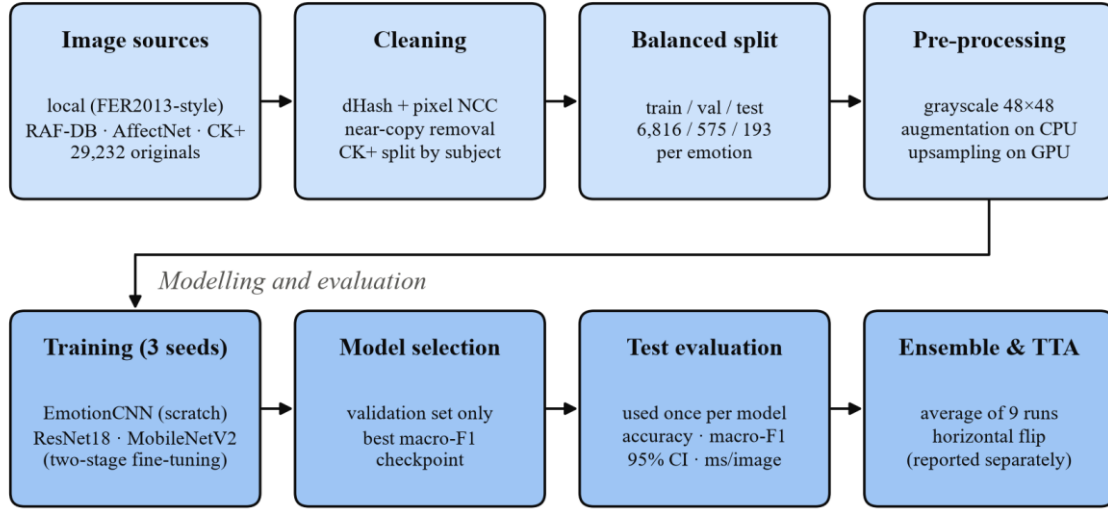
The FER2013 challenge [1] introduced 35,887 web-collected 48×48 grayscale faces; its organizers estimated human accuracy at about $65 \pm 5\%$, reflecting both ambiguous expressions and noisy labels. Convolutional networks quickly surpassed this level: a tuned VGG-style network reaches 73.3% on the FER2013 test set [18], and ensembles of several CNNs reach about 75% [19]. Microsoft's FER+ [5] relabeled every FER2013 image with ten annotators, and training on the resulting majority or distribution labels gives markedly higher accuracy, which shows how much label noise limits FER2013.

Larger in-the-wild datasets followed. RAF-DB [2] provides about 15,000 crowd-annotated color images, on which transformer-based models such as POSTER++ [20] reach about 92% accuracy. AffectNet [3] contains more than one million web images; on its balanced seven-class validation set the best methods reach only about 67%, again illustrating how hard spontaneous expressions are. Most recent methods use high-resolution color images and large backbones pretrained on face recognition; our study instead targets small grayscale faces and models that can run in real time on a laptop.

Chapter 3. SOLUTIONS/ METHODS

Fig. 1 summarizes the proposed pipeline. In the data-preparation stage, the four image sources are cleaned of cross-split near copies, split into exactly balanced training, validation and test sets, and pre-processed. In the modelling stage, the three networks are trained with three seeds each, the best checkpoint is selected on the validation set only, and every final model is evaluated once on the test set; ensembles and test-time augmentation are reported separately. The final phase will deploy the selected model in a real-time webcam application.

Data preparation



Final phase: real-time webcam application with the selected model

Fig. 1 Block diagram of the proposed pipeline

3.1. Dataset

The dataset combines four sources of 48×48 grayscale face crops (Table. 1). All images are already aligned face crops, so no face detector is needed for training. Fig. 2 shows typical examples of each emotion.

Table. 1 IMAGE SOURCES (ORIGINAL IMAGES, ALL SPLITS)

Source	Type	Images	Emotions covered
local (FER2013-style)	In the wild, 48×48, noisy labels	15,198	all 7
RAF-DB [2]	In the wild, crowd-annotated	9,979	all 7
AffectNet [3]	In the wild	3,133	angry, disgust, fear
CK+ [4]	Posed, laboratory, 106 subjects	922	all except neutral
Total		29,232	



Fig. 2 Example test images of each emotion and their source

Cleaning and splitting. A first inspection revealed validation and test images that were near copies of training images (the same photograph re-cropped or re-exposed, hence a different file hash). Such leakage inflates the measured accuracy. We detected copies with a 64-bit difference hash (dHash) and confirmed them with the normalized cross-correlation of the pixels; 282 copies were moved out of the validation and test splits (not deleted), and 306 further images were trimmed at random so that every emotion keeps the same count. CK+ images were assigned to the splits *by subject* (84, 16 and 6 people for training, validation and test), so that no person appears in two splits. After cleaning, no exact, mirrored or near copy remains across splits.

Table. 2 TRAIN / VALIDATION / TEST SPLIT

Split	Original images	Augmented copies	Total	Per emotion
Train	23,856	23,856	47,712	6,816
Validation	4,025	–	4,025	575
Test	1,351	–	1,351	193

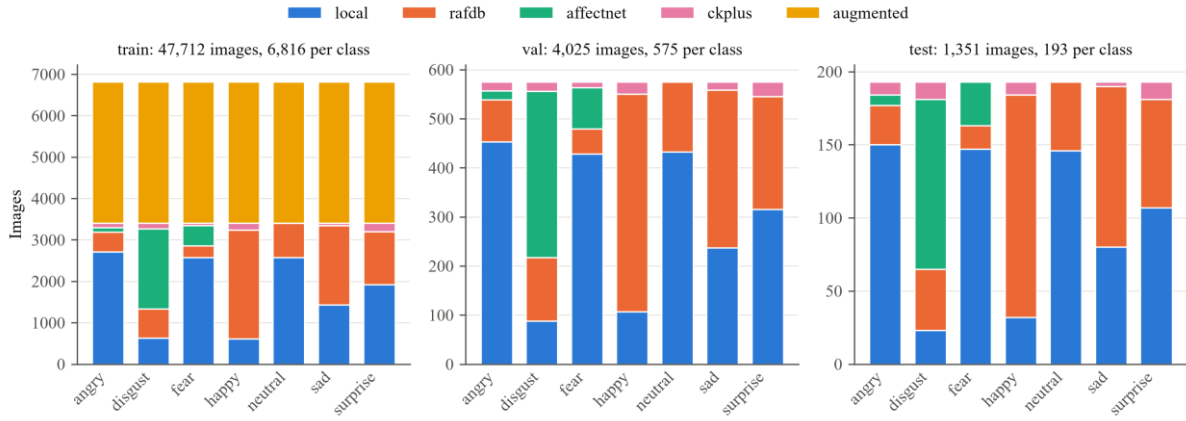


Fig. 3 Number of images per emotion and source in each split

Every split is exactly balanced (Fig. 3), so no emotion dominates the accuracy and no class weighting is needed. The validation split is used for all choices of the training recipe and of the checkpoint; the test split is used only once per final model.

Pre-processing and augmentation. Images are kept in grayscale. During training each image goes through a random resized crop (scale 0.85–1.0), horizontal flip ($p = 0.5$), rotation of up to 10° , brightness and contrast jitter of $\pm 30\%$ and random erasing ($p = 0.25$). Each training original also has one offline-augmented copy. The transforms run on the CPU at 48×48 ; each batch is then upsampled on the GPU to the model input size (96×96 for EmotionCNN, 224×224 for the pretrained networks).

3.2. Network Architecture

EmotionCNN (Fig. 4, Table. 3) is a compact VGG-style network designed for 48×48 faces. Four blocks of two 3×3 convolutions with batch normalization and ReLU extract increasingly abstract features; the first three blocks halve the resolution with max pooling, while the last block uses dilated convolutions ($r = 2$) to enlarge the receptive field to the whole face without further down-sampling. Spatial dropout grows from 0.10 to 0.25 with depth, and global average pooling followed by dropout 0.5 and a linear layer produces the seven class scores. Convolutions are initialized with Kaiming initialization [17].

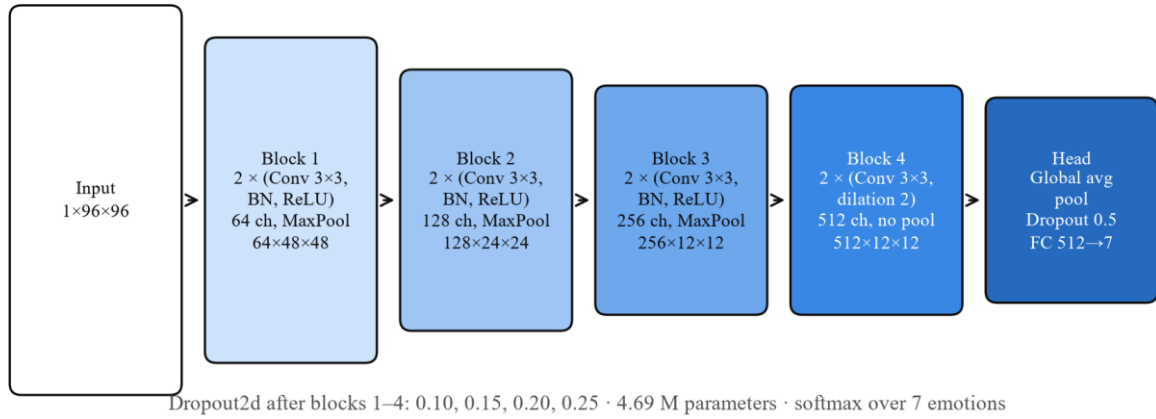


Fig. 4 Architecture of EmotionCNN

Table. 3 DETAILED STRUCTURE OF EMOTIONCNN

Stage	Layers	Output size
Input	Grayscale face, upsampled $48 \rightarrow 96$	$1 \times 96 \times 96$
Block 1	$[\text{Conv } 3 \times 3 (64), \text{BN, ReLU}] \times 2, \text{MaxPool } 2, \text{Dropout2d } 0.10$	$64 \times 48 \times 48$
Block 2	$[\text{Conv } 3 \times 3 (128), \text{BN, ReLU}] \times 2, \text{MaxPool } 2, \text{Dropout2d } 0.15$	$128 \times 24 \times 24$
Block 3	$[\text{Conv } 3 \times 3 (256), \text{BN, ReLU}] \times 2, \text{MaxPool } 2, \text{Dropout2d } 0.20$	$256 \times 12 \times 12$
Block 4	$[\text{Conv } 3 \times 3 (512), \text{dilation } 2, \text{BN, ReLU}] \times 2, \text{Dropout2d } 0.25$	$512 \times 12 \times 12$
Head	Global average pooling, Dropout 0.5, FC $512 \rightarrow 7$, softmax	7
Parameters	4,689,735 trainable (17.9 MB)	

Transfer-learning baselines. ResNet18 [11] and MobileNetV2 [12] are initialized with ImageNet weights; their classifier is replaced by dropout 0.5 and a seven-way linear layer, and the grayscale input is replicated to three channels. They are fine-tuned in two stages: stage A trains only the new classifier for 3 epochs, then stage B fine-tunes the whole network for 20 epochs with layer-wise learning rates, small for the early generic layers and larger for the last block and the classifier (Table. 4). Fine-tuning every layer lets the low-level filters adapt to small grayscale faces, which differ strongly from ImageNet photographs.

Table. 4 THE THREE MODELS

Model	Pre-training	Input	Parameters	Trained layers
EmotionCNN	None (from scratch)	$1 \times 96 \times 96$	4.69 M	All layers, 40 epochs
ResNet18	ImageNet	$3 \times 224 \times 224$	11.18 M	Classifier (3 epochs), then all layers (20 epochs)
MobileNetV2	ImageNet	$3 \times 224 \times 224$	2.23 M	Classifier (3 epochs), then all layers (20 epochs)

3.3. Experiments

Implementation details. All models share one PyTorch training loop driven by configuration files (Table. 5). Training ran on a laptop with an NVIDIA GeForce RTX 3050 Laptop GPU (4 GB), an Intel Core i5-11400H CPU and 16 GB of RAM, using mixed precision. One run takes about 50 minutes for EmotionCNN, 37 minutes for ResNet18 and 39 minutes for MobileNetV2. Early stopping was disabled: every run follows its full cosine schedule and the checkpoint with the best validation macro-F1 is kept, because stopping on a noisy validation curve often ended runs before the low-learning-rate phase in which the models still improved.

Table. 5 TRAINING SETTINGS

Setting	EmotionCNN	ResNet18 / MobileNetV2
Optimizer	AdamW	AdamW
Learning rate	1×10^{-3}	Stage A 1×10^{-3} ; stage B 5×10^{-4} (classifier), 1×10^{-4} (last block), $3-5 \times 10^{-5}$ (middle), 2×10^{-5} (early layers)
Weight decay	5×10^{-4}	5×10^{-2}
Batch size	64	32
Epochs	40	3 + 20
Schedule	1-epoch linear warm-up, cosine decay to 1×10^{-6} (per stage)	same
Loss	Cross-entropy, label smoothing 0.1	same
Other	Gradient clipping 5.0, mixed precision, best validation macro-F1 checkpoint	same
Seeds	42, 123, 2024	42, 123, 2024

Evaluation protocol. We report accuracy and macro-F1 (the unweighted mean of the seven per-class F1 scores, so that no emotion is neglected) as the mean $\pm$ standard deviation over the three seeds. The 95% confidence interval of each test score is estimated with 1,000 bootstrap resamples of the test set [24]. Inference time is measured with batch size 1 (20 warm-up and 200 timed passes) on the CPU and on the GPU. Ensembles and horizontal-flip test-time augmentation (TTA) were fixed before the test set was used and are reported separately from the single models.

Chapter 4. RESULTS AND DISCUSSION

4.1. Learning Behaviour

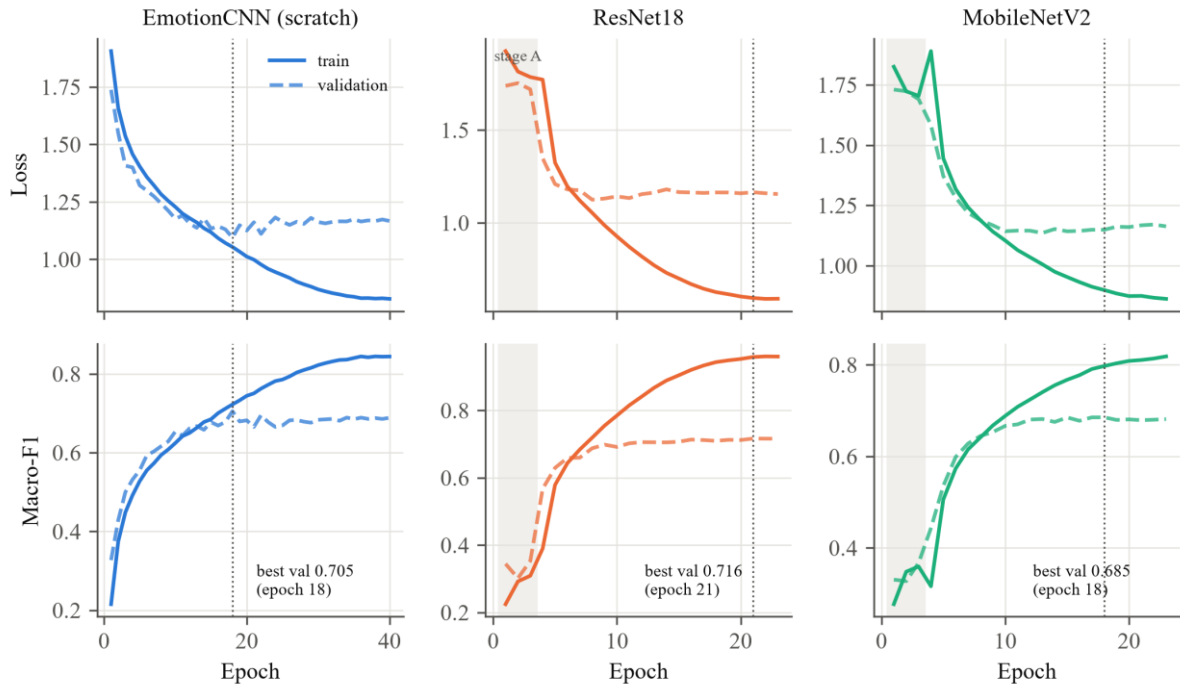


Fig. 5 Learning curves of the three models (seed 42); the dotted line marks the best validation epoch

Fig. 5 shows that all three models converge smoothly. EmotionCNN's training and validation curves stay close until about epoch 18, where it reaches its best validation macro-F1 (0.705); afterwards the training F1 keeps rising to 0.84 while the validation F1 levels off around 0.69, a mild overfitting that the best-checkpoint rule absorbs. The pretrained networks gain little in stage A (shaded): with frozen ImageNet features the validation macro-F1 stays near 0.35, which confirms that the features must be fine-tuned for small grayscale faces. In stage B, ResNet18 fits the training set almost perfectly (training F1 0.96) while its validation F1 settles near 0.71, the largest generalization gap of the three models.

4.2. Test Performance

Table. 6 TEST RESULTS (MEAN $\pm$ STD OVER 3 SEEDS)

Model	Params	Val. macro-F1	Test accuracy (%)	Test macro-F1	95% CI (macro-F1)	CPU ms/img	GPU ms/img
EmotionCNN	4.69 M	0.700	68.0 ± 1.0	0.679 ± 0.012	0.654 – 0.702	13.5	1.5
ResNet18	11.18 M	0.711	68.3 ± 1.0	0.683 ± 0.010	0.658 – 0.706	18.2	2.9
MobileNetV2	2.23 M	0.684	67.8 ± 0.9	0.678 ± 0.009	0.653 – 0.701	13.0	5.9

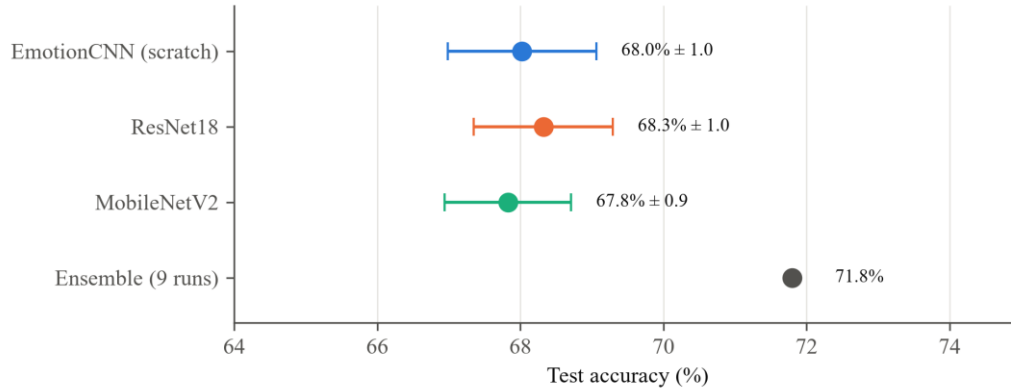


Fig. 6 Test accuracy of the three models (mean $\pm$ std over 3 seeds) and of the 9-run ensemble

The three models reach almost the same test accuracy (Table. 6, Fig. 6): the differences, below 0.5 percentage points, are smaller than the seed-to-seed standard deviation, and the bootstrap confidence intervals overlap almost entirely. The custom EmotionCNN trained from scratch therefore matches the two ImageNet-pretrained networks on this task, with 2.4 times fewer parameters than ResNet18. Test scores are about 2 points lower than validation scores, as expected, since the checkpoints were selected on the validation split. All models are fast enough for real-time use: 13–18 ms per image on the CPU, i.e. more than 50 frames per second.

4.3. Ensemble and Test-Time Augmentation

Table. 7 ENSEMBLES ON THE TEST SET (AVERAGED SOFTMAX OUTPUTS)

Ensemble	Runs	TTA	Accuracy (%)	Macro-F1
ResNet18, 3 seeds	3	yes	72.4	0.723
All models, all seeds	9	yes	71.9	0.717
All models, all seeds	9	no	71.8	0.716
EmotionCNN, 3 seeds	3	no	69.6	0.694
MobileNetV2, 3 seeds	3	yes	69.4	0.693
Single model (mean)	1	no	68.0	0.679

Averaging the softmax outputs of the nine trained models raises the test accuracy from 68.0% to 71.8% (95% CI 69.4–74.3%), because the models make partly different mistakes (Table. 7). Horizontal-flip TTA adds up to 1 more point for the pretrained models, but slightly lowers EmotionCNN (69.6% $\rightarrow$ 69.3%). These ensembles are reported separately; a real-time application would use a single model or a small ensemble.

4.4. Error Analysis

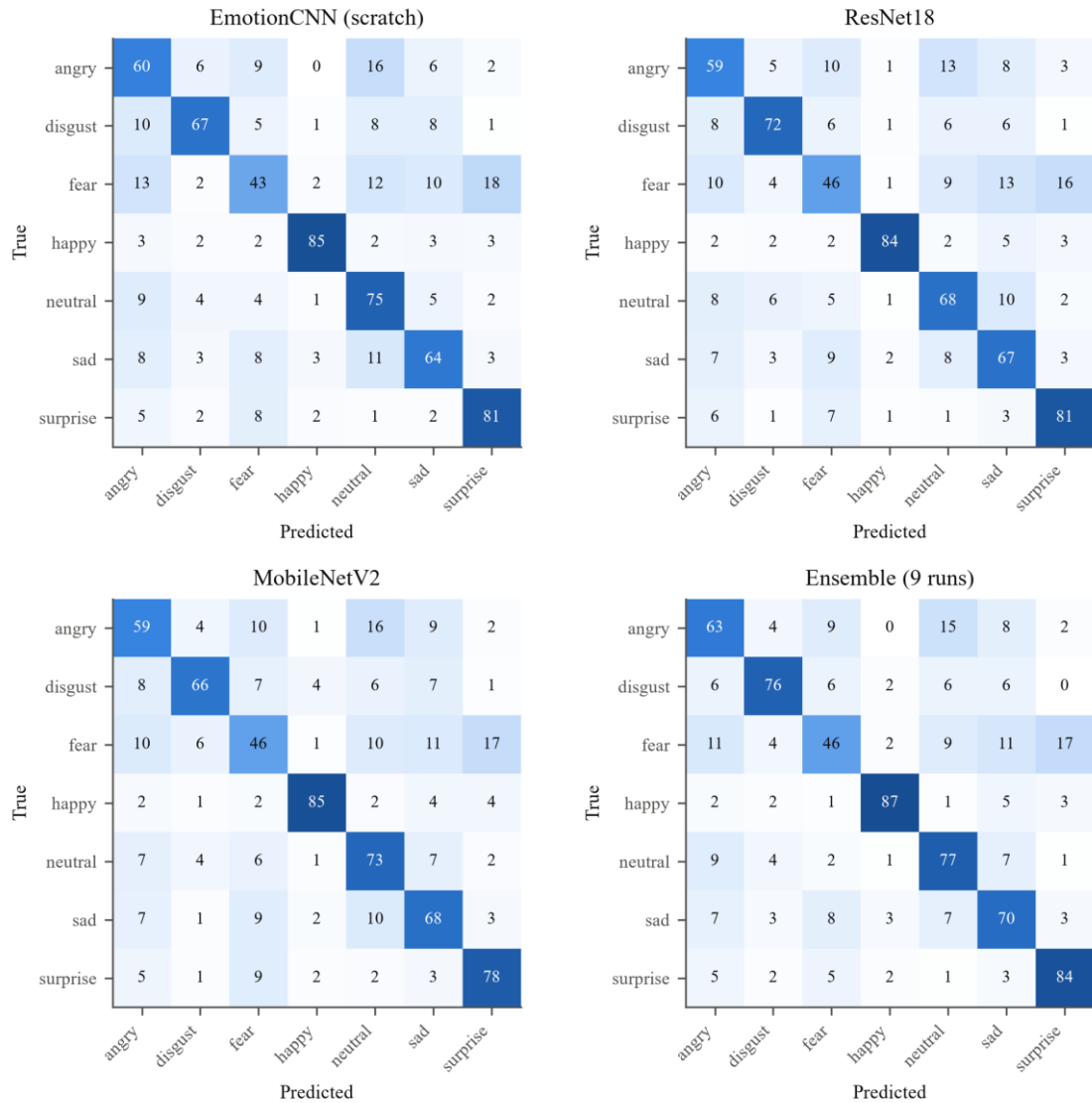


Fig. 7 Row-normalized confusion matrices on the test set (%); single models sum the three seeds

The confusion matrices (Fig. 7) show the same pattern for every model. Happy and surprise are recognized best (above 80%), while fear is the weakest emotion: only 43–46% of fear images are classified correctly, and 16–18% of them are taken for surprise, since both expressions open the eyes and the mouth. Angry is often predicted as neutral (13–16%), because a mild frown differs from a neutral face mostly in the eyebrows.

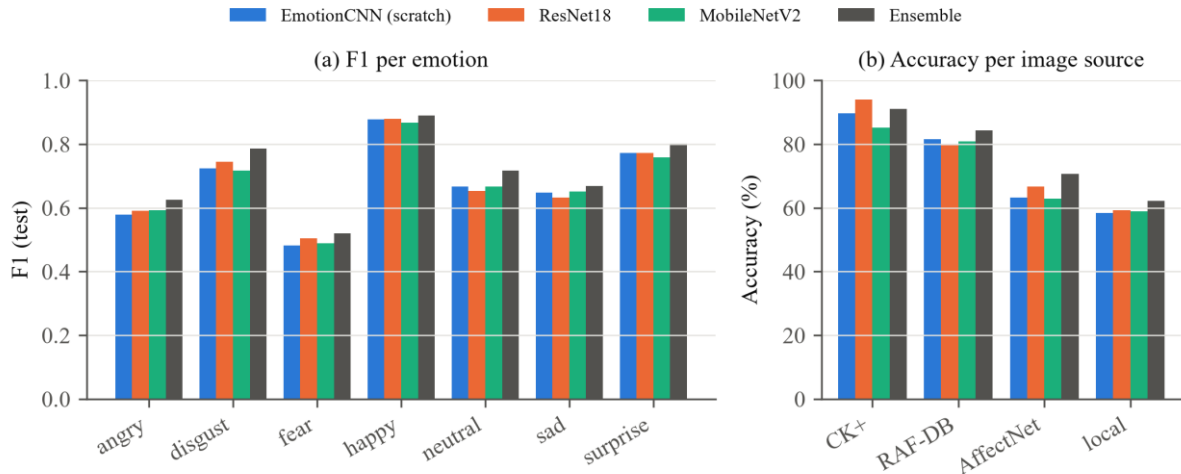


Fig. 8 F1 per emotion (a) and accuracy per image source (b) on the test set

Accuracy depends strongly on the image source (Fig. 8b): the ensemble reaches 91.1% on posed CK+ faces (45 test images only), 84.4% on RAF-DB and 70.6% on AffectNet, but only 62.2% on the FER2013-style local images, which make up half of the test set. The local images are the smallest, blurriest and most noisily labeled, so they bound the overall accuracy.

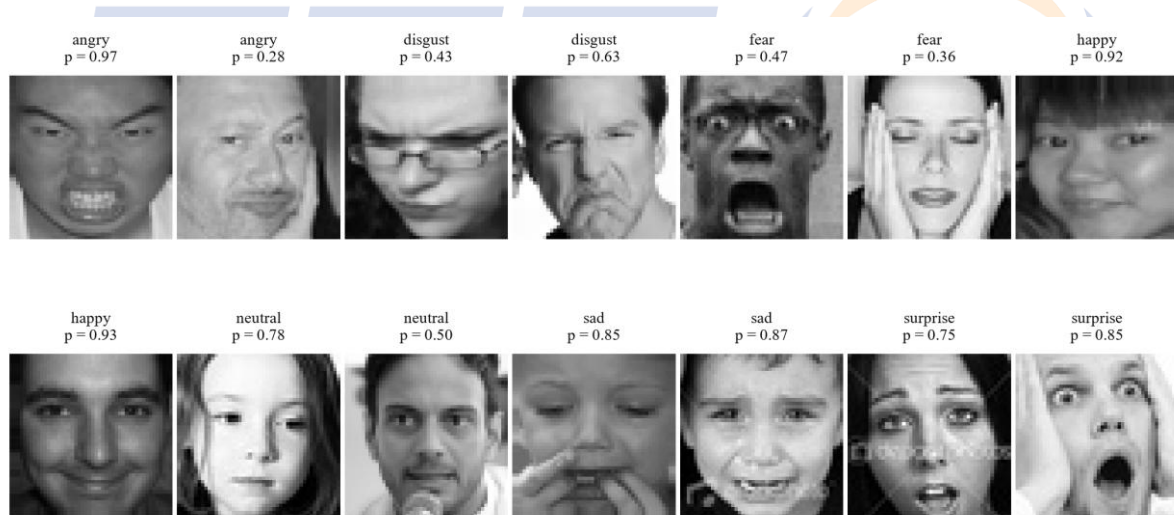


Fig. 9 Correct predictions of the ensemble with their confidence p



Fig. 10 Incorrect predictions of the ensemble (true label, predicted label, confidence p)

Correct predictions usually show a clear, typical expression, and 45% of them are made with a confidence above 0.8 (Fig. 9). Errors are mostly uncertain: 56% of the misclassified test images receive a confidence below 0.5 (Fig. 10). Several of them are genuinely ambiguous, for example a sad face crying with an open mouth predicted as fear, a surprised face with both hands over the mouth predicted as fear, or a weak smile under face paint. Some labels themselves are debatable, which is consistent with the known label noise of FER2013.

4.5. Comparison with Published Results

Table. 8 COMPARISON WITH PUBLISHED RESULTS (APPROXIMATE, DIFFERENT PROTOCOLS)

Dataset	Image type	Best published	This work (ensemble, test)
CK+ [4]	Posed, laboratory	> 95%	91.1% (45 images, 6 subjects)
RAF-DB [2]	In the wild, color	≈ 92% [20]	84.4% (grayscale 48×48)
FER2013 [1]	In the wild, 48×48 grayscale	73.3% single [18], ≈ 75% ensemble [19]; human ≈ 65% [1]	62.2% (local FER2013-style images)
AffectNet [3]	In the wild	≈ 67% (7 classes) [20]	70.6% (3 emotions only, not comparable)

Table. 8 places the results in context. The published numbers come from different protocols, image resolutions and backbones, so the comparison is only indicative. On each source our models stay within roughly 7–12 points of the best published methods while using small grayscale images and models that run in real time on a laptop. The very high accuracies often quoted for FER (above 90%) hold for posed laboratory datasets; on in-the-wild faces, and on FER2013-style images in particular, accuracies around 70% are typical.

Chapter 5. CONCLUSION AND RECOMMENDATION

5.1. Conclusion

This midterm study built a clean, exactly balanced and leakage-free seven-class FER dataset of 53,088 grayscale images from four sources, and used it to train and compare three networks under one reproducible protocol. The custom EmotionCNN, trained from scratch with batch normalization, spatial dropout and dilated convolutions, reaches $68.0 \pm 1.0\%$ test accuracy and a macro-F1 of 0.679, statistically on par with fine-tuned ResNet18 (68.3%) and MobileNetV2 (67.8%), while having 2.4 times fewer parameters than ResNet18 and running in 13.5 ms per image on a CPU. Averaging the nine trained models raises the accuracy to 71.8%.

The error analysis shows that the remaining errors are concentrated on fear (F1 0.52), which is confused with surprise, and on the FER2013-style images, where accuracy is 62% against 84–91% on RAF-DB and CK+. Most errors are low-confidence predictions on ambiguous or debatably labeled faces, which indicates that label noise, rather than model capacity, is now the main limit.

5.2. Recommendation

For the final report we recommend the following steps, in order of expected benefit:

- **Relabel the FER2013-style images with FER+ annotations [5].** Ten annotators per image allow wrong labels to be corrected and ambiguous images to be removed; this targets the largest and weakest part of the data.
- **Explain the predictions with Grad-CAM [21].** Visualizing which facial regions drive each decision will show whether the models attend to the eyes, eyebrows and mouth, and may explain the fear–surprise confusion.
- **Build the real-time webcam application.** With 13 ms per image on a CPU, EmotionCNN or MobileNetV2 should run at about 30 frames per second including face detection, with temporal smoothing of the predictions to avoid flickering labels.
- **Collect a small real-world webcam dataset.** Webcam images are not expected to raise the accuracy on the current test set, whose faces come from thousands of unknown people in the wild, but they should reduce the domain shift between the training images and the live camera, such as different lighting, sensor and face cropping [26]. We plan to record 5–10 consenting volunteers showing the seven emotions over several sessions, keep about one frame per second to avoid near-identical frames, and crop the faces with exactly the same detector and 48×48 grayscale processing as the demo. The data will first serve as a separate webcam test set, to measure how much accuracy drops in real use; MobileNetV2 will then be fine-tuned on some volunteers and evaluated on the others, so that no person appears in both training and testing.
- **Add more data for the weakest emotions,** for example additional fear and disgust images, or the JAFFE dataset once access is granted by its authors.

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